

GARDSJO, Sweden

WMO: 025460

Lat: 58.8768N

Lon: 14.3875E

Elev: 696

StdP: 14.33

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.6	10.2	2.1	6.3	7.7	6.0	7.7	12.0	19.9	32.9	17.6	32.4	4.5	60	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.1	78.1	63.4	74.9	61.9	71.7	60.8	65.9	73.5	64.1	71.5	62.4	69.0	5.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.3	89.2	68.5	61.4	83.3	66.7	59.6	78.2	65.2	31.0	73.6	29.6	71.8	28.4	69.2	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.4	13.7	12.1	2.8	82.0	5.3	3.0	-1.0	84.1	-4.1	85.9	-7.0	87.6	-10.9	89.8		
			1.7	68.5	6.0	2.1	-2.5	70.0	-6.0	71.2	-9.4	72.4	-13.7	73.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.5	26.6	27.0	32.7	42.2	51.2	57.6	61.9	59.9	52.6	43.3	35.9	30.4
	DBStd	14.08	8.15	7.88	7.48	6.32	6.76	5.49	5.30	4.68	5.06	6.22	6.74	8.50
	HDD50	3499	725	643	537	246	65	4	0	1	30	219	423	606
	HDD65	7888	1190	1063	1001	683	430	232	130	170	372	673	873	1071
	CDD50	1142	0	0	0	13	101	231	369	309	108	11	0	0
	CDD65	59	0	0	0	0	2	10	34	13	0	0	0	0
	CDH74	357	0	0	0	0	13	84	214	46	0	0	0	0
	CDH80	32	0	0	0	0	0	6	23	3	0	0	0	0
Wind	WSAvg	6.0	6.3	6.3	6.6	6.2	5.9	5.6	5.2	5.0	5.7	6.1	6.8	6.2
Precipitation	PrecAvg	28.3	2.0	1.8	1.3	1.6	1.9	2.8	3.3	3.2	2.7	2.8	2.5	2.2
	PrecMax	37.7	3.9	3.9	2.5	4.5	3.8	6.4	7.3	6.1	6.4	6.4	5.4	3.9
	PrecMin	22.8	0.3	0.4	0.2	0.3	0.5	0.2	0.9	1.0	0.8	1.0	0.7	0.5
	PrecStd	3.4	0.9	0.9	0.6	1.0	0.9	1.4	1.5	1.3	1.5	1.3	1.3	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.0	45.1	55.0	65.8	75.8	80.8	82.7	78.3	70.2	58.0	51.3	48.2
		MCWB	42.6	40.9	45.6	51.4	58.8	62.2	65.8	64.6	61.1	54.1	49.2	47.1
	2%	DB	40.8	41.3	49.2	61.7	71.4	76.1	79.4	74.6	65.9	55.7	49.1	45.5
		MCWB	39.3	39.3	42.0	49.6	56.9	60.5	64.8	63.4	58.3	52.9	47.7	44.5
	5%	DB	39.0	39.3	45.3	57.0	67.7	72.4	76.1	71.2	62.9	53.8	47.1	43.1
		MCWB	37.8	37.8	40.2	46.7	55.5	59.1	63.7	62.1	57.4	51.7	45.8	41.8
	10%	DB	37.0	37.3	42.2	52.9	63.5	68.4	72.5	68.2	60.2	52.0	45.0	41.0
		MCWB	35.9	35.9	38.3	44.3	53.0	57.5	62.2	60.6	55.5	50.2	43.9	40.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.9	42.3	46.4	54.0	62.1	64.9	68.8	68.0	62.4	55.7	50.2	47.5
		MCDB	43.6	43.3	53.4	62.7	70.8	74.0	77.7	73.4	66.8	56.8	50.7	48.1
	2%	WB	39.9	40.0	43.2	50.2	59.5	62.3	66.7	65.3	59.9	53.7	47.8	44.7
		MCDB	40.6	41.0	47.8	58.6	66.5	72.9	75.2	71.0	64.3	54.9	48.6	45.3
	5%	WB	37.9	38.2	41.2	47.7	56.7	60.6	65.0	63.4	57.9	52.1	46.0	42.1
		MCDB	38.8	39.2	44.7	55.6	64.5	69.5	73.5	68.6	61.8	53.4	47.0	42.9
	10%	WB	36.0	35.8	39.1	45.5	54.6	58.6	63.3	61.7	56.3	50.4	44.1	40.1
		MCDB	37.0	37.0	41.7	51.6	62.2	66.4	70.9	66.6	59.6	51.8	44.9	41.0
Mean Daily Temperature Range	5% DB	MDBR	6.9	8.0	11.1	15.1	16.0	15.9	15.1	13.3	11.9	9.3	6.7	6.7
		MCDBR	7.8	8.6	15.3	20.2	20.4	20.7	20.0	16.7	14.6	9.5	8.4	7.9
		MCWBR	7.8	7.0	9.9	11.7	11.2	10.4	9.3	8.6	8.5	7.0	8.0	8.1
	5% WB	MCDBR	7.8	8.2	13.7	18.6	18.3	18.7	17.8	14.5	13.5	8.5	8.2	8.0
		MCWBR	7.9	7.1	9.6	11.2	11.0	10.4	9.2	8.8	8.6	7.2	8.0	8.4
Clear-Sky Solar Irradiance	taub	0.267	0.304	0.326	0.363	0.360	0.350	0.371	0.363	0.334	0.325	0.291	0.249	
	taud	2.231	2.311	2.384	2.354	2.422	2.469	2.425	2.443	2.513	2.459	2.305	2.157	
	Ebn at Noon	178	229	260	267	277	281	271	265	256	221	167	140	
	Edh at Noon	14	22	27	33	33	32	33	30	24	19	13	10	
All-Sky Solar Radiation	RadAvg	105	298	782	1250	1611	1815	1629	1288	868	391	135	72	
	RadStd	16	51	88	144	145	99	192	136	115	66	27	7	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (59 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	-236	-295	N/A	N/A

Nomenclature: See separate page